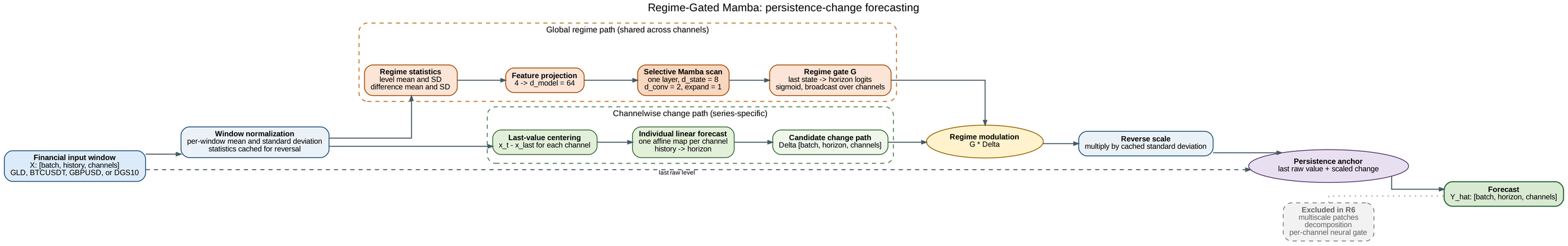
Regime-Gated Mamba: persistence-change forecasting
Global regime path (shared across channels)
Regime statistics
level mean and SD
difference mean and SD
Feature projection
4 -> d_model = 64
Selective Mamba scan
one layer, d_state = 8
d_conv = 2, expand = 1
Regime gate G
last state -> horizon logits
sigmoid, broadcast over channels
Channelwise change path (series-specific)
Financial input window
X: [batch, history, channels]
GLD, BTCUSDT, GBPUSD, or DGS10
Window normalization
per-window mean and standard deviation
statistics cached for reversal
Last-value centering
x_t - x_last for each channel
Individual linear forecast
one affine map per channel
history -> horizon
Candidate change path
Delta [batch, horizon, channels]
Regime modulation
G * Delta
Reverse scale
multiply by cached standard deviation
Persistence anchor
last raw value + scaled change
last raw level
Forecast
Y_hat: [batch, horizon, channels]
Excluded in R6
multiscale patches
decomposition
per-channel neural gate